

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The number of bacteria in a liquid medium doubles every day. There are **44,000** bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

Answer

- A. $y = \frac{1}{2}(44,000)^t$
- B. $y = 2(44,000)^t$
- C. $y = 44,000(\frac{1}{2})^t$
- D. $y = 44,000(2)^t$

Correct Answer: D

Rationale

Choice D is correct. Since the number of bacteria doubles every day, the relationship between t and y can be represented by an exponential equation of the form $y = a(b)^t$, where a is the number of bacteria at the start of the observation and the number of bacteria increases by a factor of b every day. It's given that there are **44,000** bacteria at the start of the observation. Therefore, $a = 44,000$. It's also given that the number of bacteria doubles, or increases by a factor of **2**, every day. Therefore, $b = 2$. Substituting **44,000** for a and **2** for b in the equation $y = a(b)^t$ yields $y = 44,000(2)^t$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the number of bacteria is decreasing by half, not doubling, every day.